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Summer Internship, June-July,2025- CODMAV
Review-1

Project Title : Post Quantum Cryptography in federated learning for credit risk analysis

Project ID :10

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Outline

- Problem Definition
- Objectives & Feasibility Study
- Literature Survey
- Proposed Approach
- System Architecture
- Tools & Technologies
- Timeline & Milestones

Problem Statement

Well-defined problem statement

Federated learning offers a compelling solution by enabling multiple financial institutions to collaboratively train machine learning models without sharing raw data, thus preserving privacy while leveraging distributed datasets for improved model performance.
Credit risk analysis requires a varying dataset base from a number of banks allowing it to truly recognise the risk factor of an applicant for a credit, and hence Federated Learning fits perfectly for the use case of banks as edge devices.

This work addresses the growing threat of quantum computing by integrating Post-Quantum Cryptography (PQC)—specifically lattice-based and code-based approaches—into the FL process for credit risk analysis. By securing communication and model updates with quantum-resistant algorithms, our framework ensures long-term data privacy and system integrity. The significance of this research lies in creating a secure, scalable, and future-proof FL infrastructure for the financial industry, bridging the gap between privacy-preserving analytics and quantum-resilient security.

Why was the problem selected?

Banks for years have struggled with including AI augmentations into their workflow as the idea of pooling their user confidential and even bank-wise sensitive data into a single repository for model Training was a direct data privacy leak and a competitive disadvantage.
Credit Loan approvals in India face a high defaulting rate, with no previous extensive credit history to prove it.
Another problem arises with the ongoing research in Quantum computational and decryption field, wherein standard state-of-the-art encryptions like RSA are being theorized of vulnerabilities with the rise of QC, hence a possibility to address these two problems.
An AI model using RFC's to train on the local repositories of data to understand the approval ratings of a person, while still maintaining the new standard of quantum encryption using Post Quantum Cryptography (PQC) encryption would be of significant application in the financial market.

Justifying the novelty

Here we propose the intersection of three novel fields, Integrating the data driven privacy aspects of Federated Learning with the security of PQC to address the problem of credit risk in finance for banking.
We aim to have a robust system to identify the aspects of a credit fraud by training on local repositories using FL and addressing the problem of malicious parameter passing using PQC.

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Literature Survey-Key Insights

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Title	Author(s)	Year	Approach	Dataset / Tools	Key Findings	Gap Identified
PQS-BFL: Post-Quantum Secure Blockchain-based Federated Learning	Commey & Crosby	2025	FL + blockchain + lattice-based PQC (Dilithium signatures)	MNIST, SVHN, HAR	Achieved fast PQC signing (~0.65 ms) and maintained accuracy (>98.8% on MNIST); blockchain added minor overhead (arxiv.org)	Focuses on authentication; lacks model privacy protection, no finance-specific use
PQBFL: Post-Quantum Blockchain-based Protocol for Federated Learning	GHaravi, Granjal & Monteiro	2025	Secure FL with PQC + blockchain + forward secrecy	Medical-style FL setup	Introduced ratcheting for forward secrecy in iterative FL	Focuses on secure channel and identity privacy; no aggregation/model-level PQC or scalability testing
Enhancing Quantum Security over FL via PQC	Li, Chen & Liu	2024	Signature-centric PQC in FL (Dilithium, Falcon, SPHINCS+)	General FL tasks	Found Dilithium best balanced performance vs. security	Limited to authentication; lacks use case specificity and scalability
Performance Analysis of PQC-Enabled Blockchain FL	Gurung, Pokhrel & Li	2023	Hybrid stateless (Dilithium, Falcon) + stateful (XMSS) PQC + blockchain	Simulated BFL	Demonstrated practical security and performance trade-offs; used VRFs for selection	Still blockchain-focused; no financial dataset or optimization for real deployment
Secure Communication Model for Quantum FL: A PQC Framework	Gurung, Pokhrel & Li	2023	PQC-secured FL (dynamic server selection, convergence analysis)	Public implementation	Proposed dynamic server selection with convergence guarantees	Lacks comprehensive evaluation, real-world application, or resource optimization

Existing solution compared or evaluated
Limitations Observed

Approach	Limitations
Standard Federated Learning	Vulnerable to quantum attacks due to use of classical cryptography (RSA, ECC)
FL with Homomorphic Encryption	High computational and memory overhead; not post-quantum secure
Blockchain-based PQC-FL	Focus is on authentication, not model privacy or financial applications
Signature-Centric PQC in FL	Limited scalability testing; does not address credit risk modeling or resource allocation

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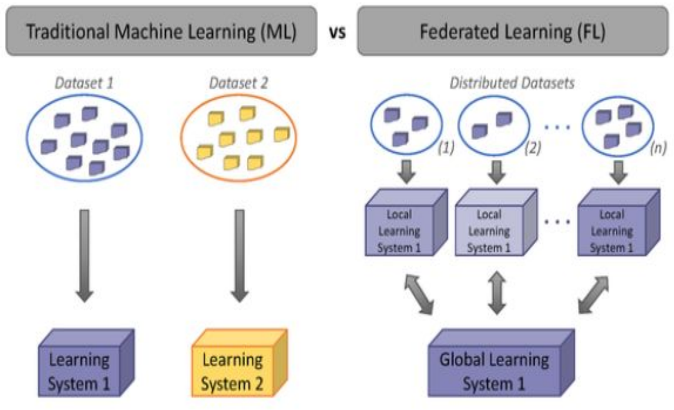
Federated learning - Proposed Approach

1. **Initialize Global Model**
   The central server creates an initial machine learning model.

2. **Distribute Model to Clients**
   The server sends the global model to selected client devices.

3. **Local Training on Clients**
   Each client trains the model on its private data.
   No raw data is shared.

4. **Send Model Updates to Server**
   Clients send only weight updates or gradients.



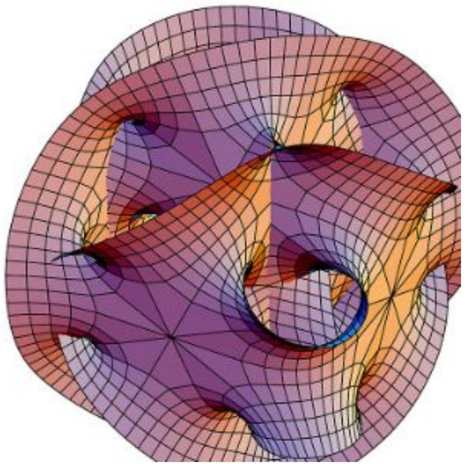
5. **Aggregate Updates at Server**
   Server uses algorithms like **Aggregation** to combine updates and improve the global model.

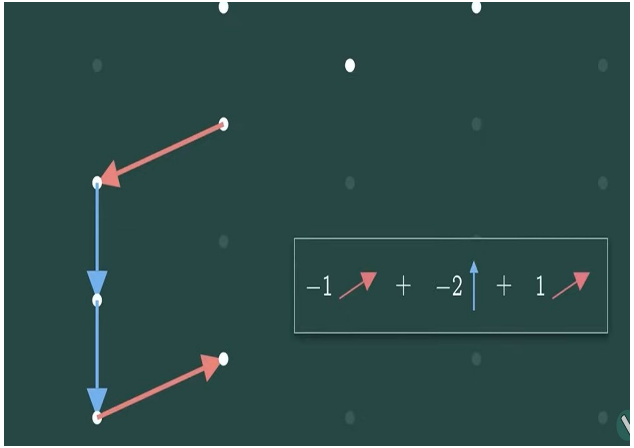
6. **Repeat Rounds**
   Steps 2–5 are repeated until the model reaches acceptable performance.

Post Quantum Cryptography - Proposed Approach

IDEA BEHIND QUANTUM CRYPTOGRAPHY

1.Post-quantum cryptography is based on **mathematical problems that remain hard to solve even for quantum computers** (like Shor's or Grover's algorithms), ensuring future-proof security for encryption, signatures, and key exchange.





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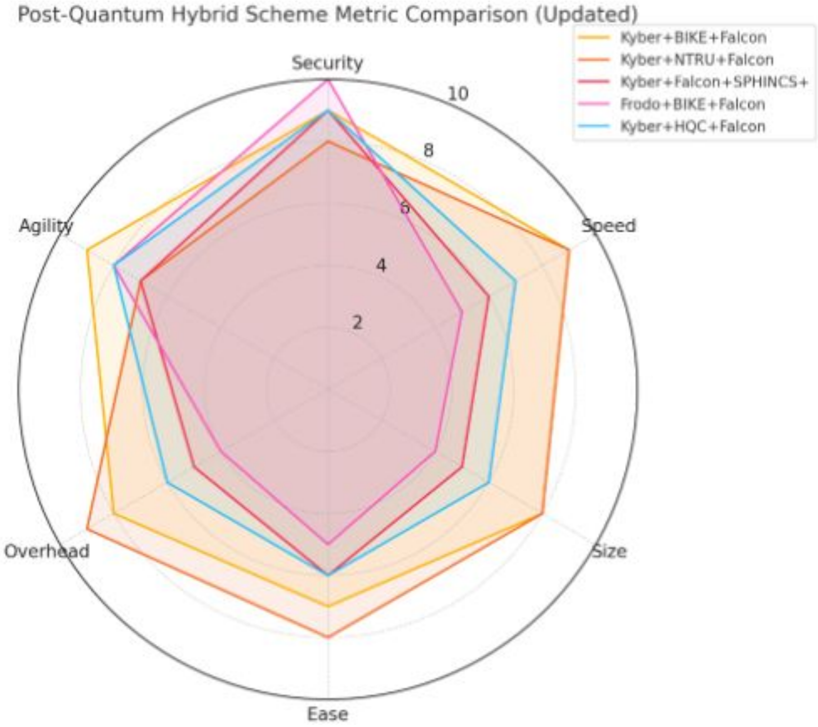
Post Quantum Cryptography - Proposed Approach



Key Exchange – Kyber / BIKE

- Bob sends a **Kyber or BIKE public key** to Alice.
- Alice uses it to generate a **shared secret**.
- → Enables **quantum-safe encryption** for communication.

Digital Signature – Falcon

- Alice signs her message using **Falcon**.
- The signature ensures the message is **authentic** and **untampered**.

Verification (Bob)

- Bob decrypts the message using the shared key.
- He verifies the **Falcon signature** to ensure it came from Alice.

•Key innovations

1. End-to-End Post-Quantum Secure Federated Learning: Not just communication, but the entire FL pipeline (key exchange, model updates, aggregation) is secured against quantum threats.

2. Dual-Scheme PQC Implementation: Combines lattice-based and code-based cryptography in one system, offering a unique comparison and enhanced flexibility.

3. Domain-Specific Application: First of its kind (to our knowledge) to apply PQC-integrated FL to credit risk analysis, supporting regulatory and privacy requirements in finance.

4. Scalable and Resource-Aware Framework: Designed to be scalable to 100+ clients, with active attention to resource constraints and deployment feasibility.

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Possible shortcomings or challenges

1.increased computational overhead and large key sizes in PQC.

2.Ensuring efficient resource usage across diverse client hardware.

3.Balancing security vs. model accuracy under PQC constraints.

4.Managing heterogeneous clients and network conditions at scale.

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System Architecture

- ## High-level architecture

```
+----------------------+   ...   +---------------------------+   ...   +---------------------+
|  Client Node 1       |         |      Client Node N        |         |  Client Node        |
| (Bank/Branch)        |         | (Bank/Branch)             |         | (Fintech Partner)   |
|----------------------|         |---------------------------|         |---------------------|
| Local Credit Data    |         | Local Credit Data         |         | Local Credit Data   |
| Preprocessing        |         | Preprocessing             |         | Preprocessing       |
| Local Model Train    |         | Local Model Train         |         | Local Model Train   |
| PQC Encryption       |         | PQC Encryption            |         | PQC Encryption      |
+---------|------------+         +------------|--------------+         +-----------|---------+
          |                                   |                                    |
          |------------ Secure PQC Channels (e.g., Kyber, McEliece) ------------->|
                                      ↓
                          +-------------------------------+
                          |       Aggregation Server      |
                          | (Secure Model Aggregator)     |
                          |-------------------------------|
                          | PQC Decryption & Verification |
                          | Secure Aggregation of Models  |
                          | Global Model Update           |
                          +---------------|---------------+
                                          |
                       Global Model (Credit Scoring)
                                          ↓
                        +-------------------------------+
                        |      Output/Applications      |
                        |-------------------------------|
                        | Credit Risk Prediction        |
                        | Model Performance Reports     |
                        | Regulatory Compliance Logs    |
                        +-------------------------------+
```

•Components/modules and tools used

1. Inputs
Local Credit Data: Income, repayment history, credit score, default status, etc. (stored at each client site).

PQC Public Keys: Used for secure communication and model encryption.

2. Processes
Local Model Training: Clients train models on their private credit data.

PQC Encryption (Client-Side): Updates are encrypted using lattice/code-based PQC (e.g., CRYSTALS-Kyber, McEliece).

Secure Transmission: Encrypted model updates are sent to the central server via PQC-secured channels.

Aggregation Server: Decrypts updates, verifies authenticity, and aggregates them into a global model.

Global Model Update: Shared with all clients in the next round.

3. Outputs
Federated Credit Risk Model: Predicts creditworthiness without centralized data.

Model Metrics: Accuracy, AUC-ROC, convergence reports.

Secure Logs: For auditability and regulatory compliance.

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•Milestones Achieved and timeline

Week 1 - Researched and outlined a problem statement to a financial problem.
Week 2 - Literature Survey onto existing solutions and compared with our proposed solution.
Week 3 - Outlining performance and general improvements and start with a prototype
Week 4 - Start with the base traditional model and see performance with PQC overhead
Week 5 - Start implementing a full scale solution and aim for resource usage minimization
Week 6 - Analyze results and compare with classical and PQC algorithms
Week 7 - Present a fully working model

Addressed the problem of finding a suitable dataset for Indian markets and augmenting the dataset to reach a good volume (1 M)
Compared the accuracies between FL and centralised models on a sample dataset to test experiments of increasing clients and epochs and rounds run.

Expected Deliverables

- Publication
- Proof of Concept(software Prototype)

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•References

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Thank
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